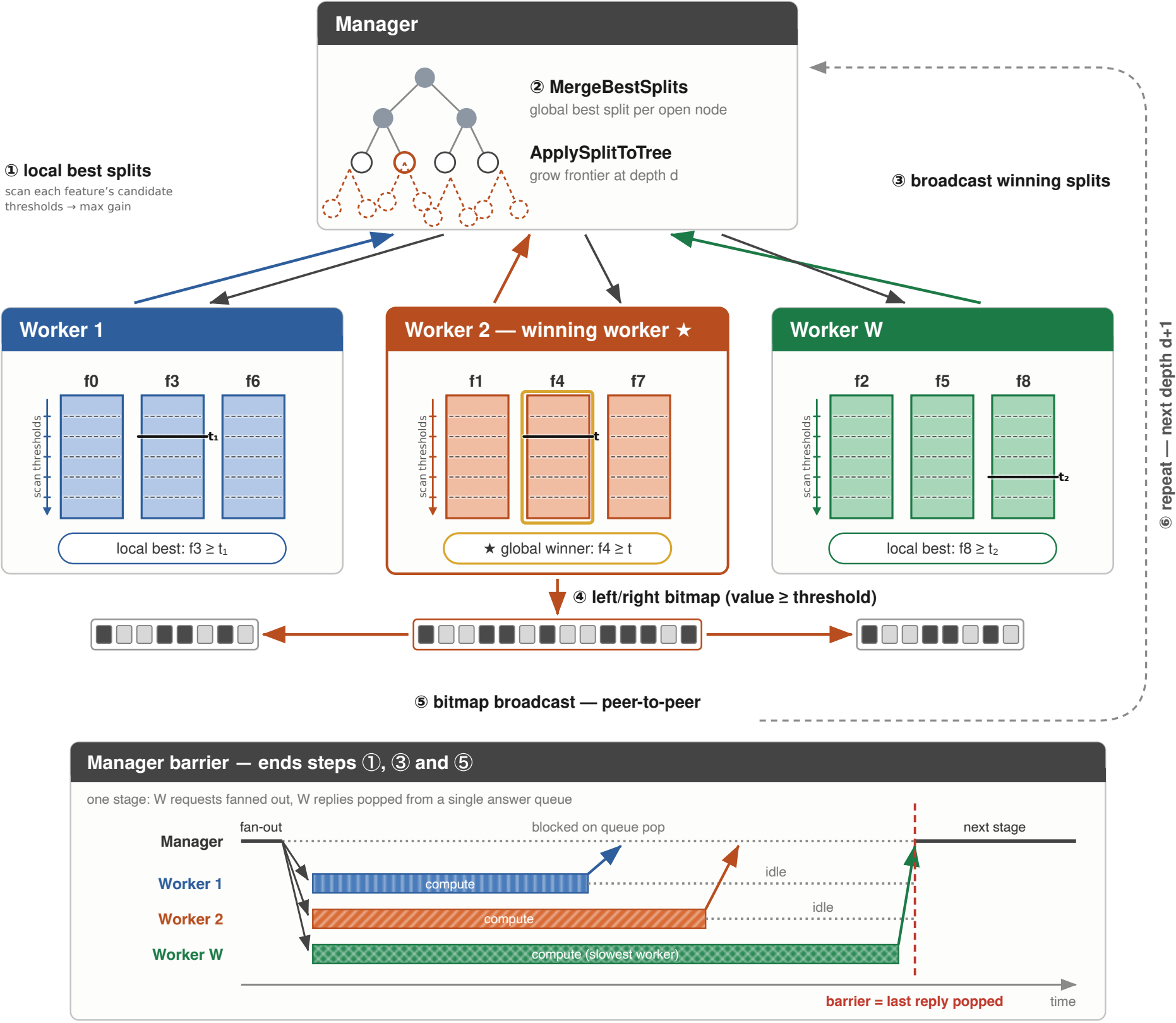

Manager
② MergeBestSplits
global best split per open node
ApplySplitToTree
grow frontier at depth d
① local best splits
scan each feature's candidate thresholds → max gain
③ broadcast winning splits
⑥ repeat — next depth d+1
Worker 1
f0
f3
f6
scan thresholds
t₁
local best: f3 ≥ t₁
Worker 2 — winning worker ★
f1
f4
f7
t
★ global winner: f4 ≥ t
Worker W
f2
f5
f8
t₂
local best: f8 ≥ t₂
④ left/right bitmap (value ≥ threshold)
⑤ bitmap broadcast — peer-to-peer
Manager barrier — ends steps ①, ③ and ⑤
one stage: W requests fanned out, W replies popped from a single answer queue
Manager
Worker 1
Worker 2
Worker W
fan-out
blocked on queue pop
next stage
compute
idle
compute
idle
compute (slowest worker)
barrier = last reply popped
time